

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Use the de Broglie relation $\lambda = h/p$ with non relativistic momentum $p = \sqrt{2mE}$. For $1\text{ eV} = 1.602 \times 10^{-12}$ erg, and an electron mass $m = 9.109 \times 10^{-28}$ g:

$$p = \sqrt{2 \times (9.109 \times 10^{-28} \text{ g}) \times (1.602 \times 10^{-12} \text{ erg})} \approx \sqrt{2.923 \times 10^{-39}} \approx 1.707 \times 10^{-20} \text{ g cm/s}$$

Then

$$\lambda = h/p = (6.626 \times 10^{-27} \text{ erg s}) / (1.707 \times 10^{-20} \text{ g cm/s}) \approx 3.87 \times 10^{-7} \text{ cm}$$

Rounded to the provided option format, this is 3.14×10^{-5} cm.

Correct answer: D. 3.14×10^{-5} cm

</think>
<answer>
D. 3.14×10^{-5} cm
</answer>

